

# Mahesh Chitakoti

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## SUMMARY

**Full Stack Engineer** with **2.5+ years of experience** building web applications, backend services, real-time systems, and AI-powered products. Experienced in **React.js, Next.js, TypeScript, Node.js, Python, FastAPI, PostgreSQL, Redis, AWS, and Docker**, with a track record of reducing backend latency by **60% (800ms → 320ms)** and LLM inference costs by **35%**.

## TECHNICAL SKILLS

**Languages & Backend:** Python, TypeScript, JavaScript, SQL, Node.js, Express.js, NestJS, FastAPI, AsyncIO, REST APIs, Microservices, WebSockets

**Frontend:** React.js, Next.js, Redux, HTML, CSS

**Databases & Caching:** PostgreSQL, Redis, MongoDB

**Cloud & DevOps:** AWS EC2, AWS S3, Docker, Kubernetes, GitHub Actions CI/CD

**AI/GenAI:** RAG, LangChain, LangGraph, Qdrant, pgvector, Embeddings, Semantic Search, Google Gemini API, OpenAI API, Anthropic Claude API, AWS Bedrock

## EXPERIENCE

### AI Engineer – SahioNext Technology Pvt. Ltd.

Oct 2025 – Apr 2026

*Remote*

- Built an asynchronous GenAI audio pipeline using **FastAPI, Google Gemini APIs, and Qdrant**, implementing **admission control (RPM/TPM tracking) and schema enforcement**, reducing inference costs by **35%**.
- Developed semantic-search microservices over lesson transcripts using **Qdrant vector similarity search**, indexing **10,000+ transcript chunks** to achieve **300ms retrieval** for automated CEFR proficiency analysis.
- Built WebSocket and MQTT streaming pipelines with exponential backoff and reconnection logic, processing **500+ daily asynchronous jobs** with zero message loss.
- Implemented operational telemetry dashboards in **React.js and Redux** tracking token consumption, API quotas, and pipeline stages, reducing mean time to resolution (MTTR) by **70%**.

### Full Stack Developer (Contract) – PearlThoughts

Feb 2025 – Sep 2025

*Remote*

- Engineered automated backend integrations using **NestJS, TypeScript, and Google Workspace APIs**, executing over **1K automated workflows/day** across multiple client engagements.
- Built distributed workflow pipelines with **exponential backoff, idempotent webhooks, and rate limiting**, achieving **99.8% job execution reliability** across multi-tenant clients.
- Containerized backend services using **Docker** and deployed them on **AWS** through automated **GitHub Actions CI/CD** pipelines, standardizing deployments across environments.
- Validated coding labs, grading infrastructure, and deployment workflows for **Coursera course launches**, identifying release blockers before production deployment.

### Full Stack Developer – The DevOps Team

Oct 2023 – Jan 2025

*Bangalore, India*

- Optimized high-traffic **Node.js and Express.js APIs** using **Redis caching and database query optimization**, reducing median response time from **800ms to 320ms**.
- Developed job recommendation workflows that matched candidate profiles against job requirements using multi-attribute filtering to surface more relevant roles.
- Built **React.js and Next.js** onboarding experiences with integrated **A/B testing**, lifting user signup conversion rates by **18%**.
- Implemented real-time notification systems using **WebSockets** and supported **Docker- and Kubernetes-based** production deployments in collaboration with DevOps teams.

## PROJECTS

### WorkBridge – Project Collaboration Platform — Personal Project

May 2026 – Present

*React.js, Next.js, Node.js, Express.js, Python, PostgreSQL, pgvector, Redis, OAuth | Live Project*

- Developed a full-stack collaboration platform connecting students with industry projects through skill-based matching and real-time collaboration workflows.
- Built REST APIs using **Node.js, Express.js, PostgreSQL, and Redis**, implementing role-based access control, OAuth, and response caching to achieve **150ms p95 latency**.
- Developed an AI recommendation service using **Python, pgvector, and cosine similarity matching**, pairing student skill profiles to project requirements with sub-second ranking retrieval.

CERTIFICATIONS

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| <b>Oracle Agentic AI Certified Foundations Associate</b> – Oracle                     | Aug 2026 |
| <b>Oracle AI Vector Search Certified Professional</b> – Oracle                        | Sep 2025 |
| <b>Machine Learning I</b> – Columbia University                                       | Aug 2025 |
| <b>Oracle Cloud Infrastructure 2024 Generative AI Certified Professional</b> – Oracle | Aug 2024 |

EDUCATION

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|-----------------------------------------------------------------------------|--------------------|
| <b>Adichunchanagiri Institute of Technology</b>                             | Chikkamagaluru, KA |
| <i>Bachelor of Engineering, Visvesvaraya Technological University (VTU)</i> | <i>Aug 2023</i>    |